CacaoML

An undergraduate exploration of Machine Learning in R.

allhailthetail/CacaoML

Matthew Younger

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Economics and Impact:

Why you should care…

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- Cacao's climate -> inherent economic inequality
  - < 6% of final value goes to farmers [fairtrade]
- National Security [Wikipedia]
  - 1700s British Navy 1 oz breakfasts [Oxford]
  - Troop morale boost
  - Calorie-dense emergency ration
- Expeditions [Wikipedia]
  - Amundsen 1911 [Wikipedia]
  - Space program since Apollo missions [Google Arts]
    - Low G affects olfactory senses
- Health benefits
  - Cognitive [Harvard]
  - Cardiovascular
  - #1 <- Quality matters! >70% only
  - Big chocolate "changing tastes…" [The Times]

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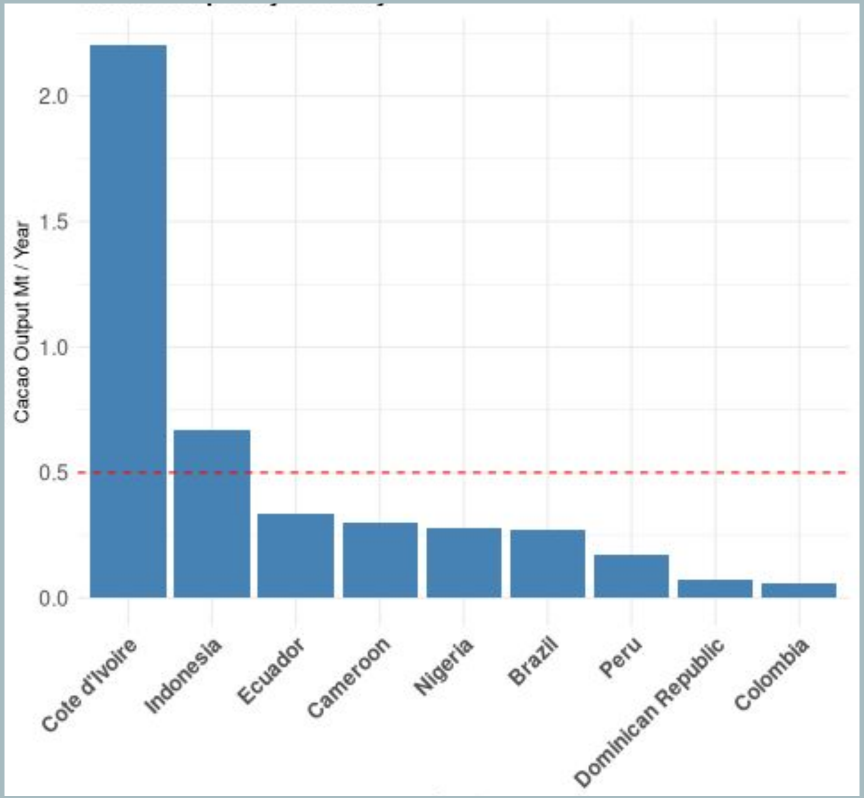

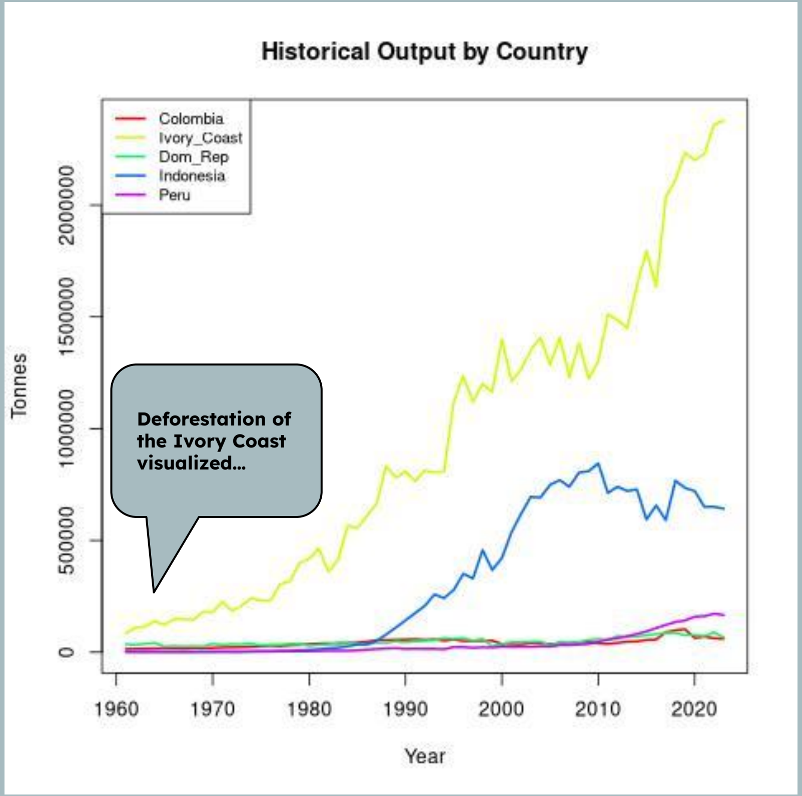

Production Overview

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Research Strategy:

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“Mind the Gap!”

- Imputation Methods:
    - **Mean**/Median/Mode
    - Spline
    - Linear
    - KNN
    - Random Forest
    - Deep Learning
    -

```
* Imputation
  - Below, I'm going to set up several different ways to do this, which I can visit later and tweak as needed
    to tune performance later:

  - ~Mean value imputation:~
    We'll stick to this for now:
    #+begin_src R
      # impute the values using mean value imputation:

      # Locate NA values for each column:
      na_tavg <- is.na(df_combined$TAVG)
      na_tmin <- is.na(df_combined$TMIN)
      na_tmax <- is.na(df_combined$TMAX)
      na_prcp <- is.na(df_combined$PRCP)
      na_kgha <- is.na(df_combined$kgha)
      na_tonnes<-is.na(df_combined$Tonnes)

      df_combined$TAVG[na_tavg] <- mean(df_combined$TAVG, na.rm = TRUE)
      df_combined$TMIN[na_tmin] <- mean(df_combined$TMIN, na.rm = TRUE)
      df_combined$TMAX[na_tmax] <- mean(df_combined$TMAX, na.rm = TRUE)
      df_combined$PRCP[na_prcp] <- mean(df_combined$PRCP, na.rm = TRUE)
      df_combined$kgha[na_kgha] <- mean(df_combined$kgha, na.rm = TRUE)
      df_combined$Tonnes[na_tonnes] <- mean(df_combined$Tonnes, na.rm = TRUE)

      summary(df_combined)
    #+end_src

  - ~Spline imputation:~
    #+begin_src R
      # impute the values with linear interpolation:
      # zoo package is helpful for this...
      install.packages("zoo")
      library(zoo)

      df_combined$TAVG <- zoo::na.spline(df_combined$TAVG)
      df_combined$TMIN <- zoo::na.spline(df_combined$TMIN)
      df_combined$TMAX <- zoo::na.spline(df_combined$TMAX)
      df_combined$PRCP <- zoo::na.spline(df_combined$PRCP)
      df_combined$kgha <- zoo::na.spline(df_combined$kgha)
      df_combined$Tonnes <- zoo::na.spline(df_combined$Tonnes)

      ## check summary again:
      summary(df_combined)
    #+end_src
```

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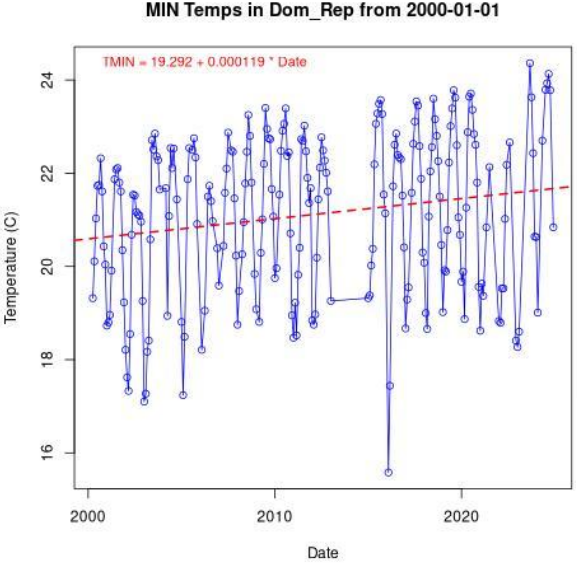
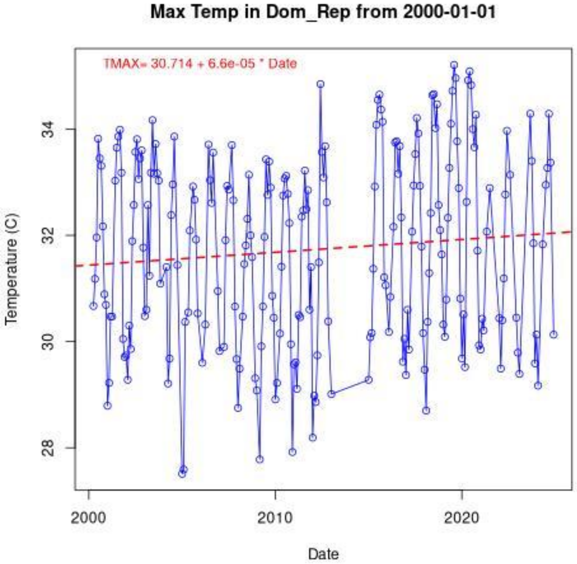
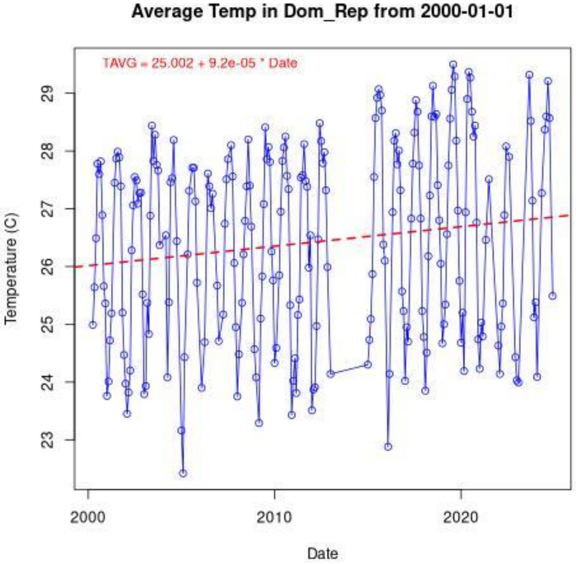
Brief “Climate Study”







🔍 Step 1: Convert to °C per Year

Since climate trends are typically expressed in **degrees Celsius per year**, we convert:

```text
9.2e-5 °C/day × 365.25 days/year ≈ 0.0336 °C/year
```

So this trend corresponds to an increase of about **+0.034 °C per year**.

🌍 Step 2: Compare to Climate Science Benchmarks

According to the **IPCC** and climate monitoring organizations like **NASA GISS** and **NOAA**, global surface temperatures have risen by approximately:

- **+0.18 to +0.20 °C per decade** over recent decades (about **+0.018 to +0.020 °C/year**),
- With warming accelerating slightly in the 21st century.

So your model’s trend of **+0.034 °C/year** is actually **on the high side** — it's nearly **double** the globally averaged trend.

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Random Forest Model:

```
- Initialize a random Forest model given the data:
  #+begin_src R :results silent
    library(randomForest)

    # Random Forest for kgha
    rf_kgha <- randomForest(kgha ~ STATION + DATE + TAVG + TMIN + TMAX + PRCP,
                            data = train_data)
  #+end_src


- Make Predictions
  #+begin_src R
    # Predictions for corrected models
    pred_kgha <- predict(rf_kgha, test_data)

    # RMSE Calculation
    rmse_kgha <- sqrt(mean((test_data$kgha - pred_kgha)^2))

    cat("Corrected RMSE for kg/ha:", rmse_kgha, "\n")

  #+end_src

  #+RESULTS:
  : Corrected RMSE for kg/ha: 57.71844
```

```
- Approximate calculations based off collected sources:
  According to Barrera, a high-performance setup is capable of packing 1,250 producing trees / ha.

  In light of these results, it seems like Random Forest has strong potential predicting production

  #+begin_src R
    # On the low end, pod weight (wet):
    # Note, this will depend a lot on what cultivar...
    pod_weight <- 0.4

    # Average pods per tree (Barrera, pg 117)
    pods_per_tree <- mean(24, 22, 22, 26, 21, 21, 22, 27)

    error_pods <- rmse_kgha / pod_weight

    error_pods / pods_per_tree -> error_trees_hectare

    print(paste("+/- trees worth of accuracy per hectare: ", round(error_trees_hectare, 2)))

  #+end_src

  #+RESULTS:
  : [1] "+/- trees worth of accuracy per hectare:  6.01"
```

This amounts to
a few trees per
acre, for the
English present...

Practical application

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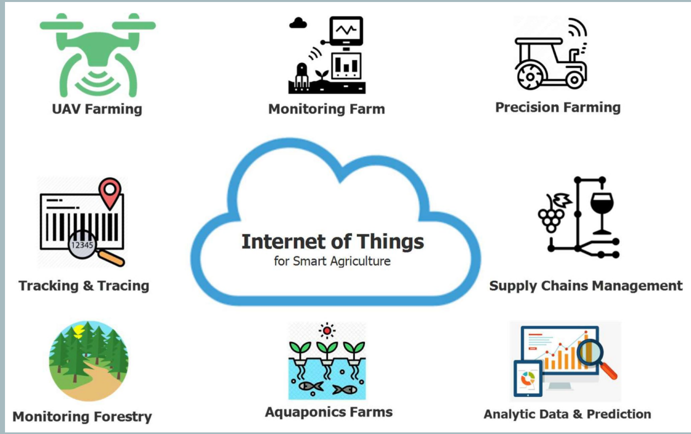
Gallery



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References:

Climate Change and Cacao Reproduction: Liam Robert Handley, The effects of climate change on the reproductive development of Theobroma cacao L., 2016, https://centaur.reading.ac.uk/72142/

Cacao Ecophysiology: Alex-Alan F. de Almeida, Raúl R. Valle, Ecophysiology of the cacao tree, 2007

Cacao Agroforestry: Ballesteros-Poss\~A\textordmasculine, William AND Navia E., Jorge Fernando AND Solarte-Guerrero, Jes\~A\textordmasculines-Geovanny, Socio-economic characterization of the traditional cacao agroforestry system (Theobroma cacao L.), 2021

V.C. Cacao Infographic: , The World's Top Cocoa Producing Countries, 2024, https://www.visualcapitalist.com/worlds-top-cocoa-producing-countries/

Socioeconomic and Technical Factors of Cacao Farming: Lina Tami-Barrera, title = {Socioeconomic and Technological Factors Influencing Technology Adoption in Cacao Farms of Two Post-Conflict Regions in Colombia},, 2021, https://etda.libraries.psu.edu/files/final_submissions/24507

Deep Learning Model for Cacao Disease Identification: Darlyn Buenaño Vera and Byron Oviedo and Washington Chiriboga Casanova and Cristian Zambrano-Vega, Deep Learning-Based Computational Model for Disease Identification in Cocoa Pods (Theobroma cacao L.), 2024, https://arxiv.org/abs/2401.01247

Expert System for Cacao Disease Diagnosis: Mohamed Othman, Hashim Hassan, Cocoa:The Construction Of Expert System For Diagnosis And Advisory Of Theobroma Cacao Plantations, 1996

Synthesis of sources made possible by...

NotebookLM

Thank You

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